

Pool Leak Detection Crash Course

A randomized technical workbook for pool circulation, leak diagnostics, shell repair, finish repair, and documentation.

Randomized version

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Safety note

This educational material is not a substitute for field training, licensing, electrical safety procedure, diving safety, excavation procedure, engineering review, or approved service procedures.

Learning Objectives

- Trace the water path through a pool circulation system.
- Separate evaporation, structural leaks, plumbing leaks, and equipment leaks.
- Explain the purpose of bucket testing, dye testing, and pressure testing.
- Distinguish concrete/gunite shell repair from plaster or pool finish repair.
- Document technical findings and repair recommendations in plain language.

Course Map

- 1. **Pool Systems 101** - Foundation, 8 min
- 2. **Understanding Water Loss** - Triage, 7 min
- 3. **The Bucket Test** - First test, 8 min
- 4. **Leak Detection Logic** - Detective work, 9 min
- 5. **Dye Testing** - Visual confirmation, 7 min
- 6. **Pressure Testing** - Plumbing isolation, 9 min
- 7. **Common Leak Locations** - Pattern recognition, 6 min
- 8. **Concrete and Guniting Shell Repair Basics** - Repair literacy, 9 min
- 9. **Pool Finish and Plaster Repair Basics** - Finish repair, 10 min
- 10. **Communicating Technical Findings** - Trust, 8 min
- 11. **Documentation and Repair Recommendations** - Company fit, 7 min
- 12. **Technical Vocabulary and Field Review** - Performance, 10 min

Six core concepts

Pools evaporate. Bucket tests separate evaporation from leaks. Pressure testing checks plumbing. Dye testing confirms visible leak points. Shell and finish defects are different repair conversations. Good diagnostics follow evidence.

Pool Systems 101

A pool is a contained body of water connected to circulation plumbing. The first technical goal is to trace the water path and identify the purpose of each major component.

Technical takeaway

The basic circulation path is: pool water is pulled through suction points, moved by the pump, cleaned by the filter, optionally heated, and returned through jets.

Key Concepts

- Skimmers pull surface water and debris into the suction side. Common leak points include cracked skimmer throats and separation from the pool shell.
- Main drains pull from the bottom. They are less common suspects than skimmers or lights, but they matter because access is harder.
- The pump is the heart of circulation. Air leaks on the suction side can show up as bubbles, poor prime, or visible issues at the lid and seals.
- Filters remove suspended debris. Cartridge, sand, and DE filters all serve the same core purpose with different media.
- Return lines push cleaned water back to the pool through fittings or jets. Return fittings and underground return plumbing can leak.

Diagnostic Notes

- Name the side of the system: suction before the pump, pressure after the pump.
- Separate visible equipment leaks from buried plumbing leaks.
- Do not diagnose from one clue. Use the clue to pick the next test.

Module Knowledge Check

M1. Which path best describes normal pool circulation?

- A. Pool to returns to filter to skimmer to pump
- B. Pool to skimmer/drain to pump to filter to heater to returns
- C. Pump to pool to skimmer to drain to heater

Answer: _____

Understanding Water Loss

Not every water drop is a leak. A technician first decides whether the loss is normal evaporation, splash-out, rain overflow, or a true leak.

Technical takeaway

Before assuming a leak, confirm abnormal water loss and account for evaporation, splash-out, wind, rain, and any autofill system.

Key Concepts

- Evaporation changes with heat, wind, humidity, sun exposure, and water features.
- An autofill can hide a leak because it replaces water while the customer sees a normal level.
- A loss of about one eighth to one quarter inch per day can be normal in many Florida conditions; more than that deserves closer testing.
- A water level that drops to a feature and then slows can point toward that feature: skimmer, light, tile line, or fitting.
- Wet soil, sinking deck areas, air bubbles, or equipment-pad puddles are clues, not final answers.

Diagnostic Notes

- Ask when the loss happens: pump on, pump off, after rain, after high wind, or constantly.
- Ask what changed recently: repair, resurfacing, new equipment, storm, heavy use.
- Look for the simplest visible issue before assuming underground plumbing.

Module Knowledge Check

M2. Why can an autofill system make a leak harder to notice?

- A.** It replaces lost water and can hide the true loss rate
- B.** It permanently seals small plumbing leaks
- C.** It makes dye testing unnecessary

Answer: _____

The Bucket Test

The bucket test compares pool water loss against a small control sample exposed to the same weather. It is one of the simplest ways to separate evaporation from leaking.

Technical takeaway

The bucket test helps distinguish evaporation from actual pool loss by comparing the pool level to a bucket of pool water over about 24 hours.

Key Concepts

- Bring the pool to normal operating level and turn off autofill.
- Put pool water in a bucket and place the bucket on a step so it sits in the same conditions.
- Mark the water level inside the bucket and the pool level outside the bucket.
- Compare after about 24 hours. Rain or a long delay can make the test unreliable.
- If the pool drops more than the bucket, the extra drop is evidence of a likely leak.

Diagnostic Notes

- The bucket is the control. Both should evaporate similarly.
- The test confirms likely leaking; it does not locate the leak.
- A positive bucket test sends the technician into isolation and location work.

Module Knowledge Check

M3. In a bucket test, what does it mean if the pool drops more than the bucket?

- A. The filter is definitely broken
- B. Likely leak beyond evaporation
- C. Normal evaporation only

Answer: _____

Leak Detection Logic

Good technicians do not guess. They run a clean diagnostic sequence: confirm the leak, separate structure from plumbing, narrow the location, verify, then recommend.

Technical takeaway

Leak detection follows a diagnostic sequence: confirm the loss, isolate the system, test likely areas, verify the finding, then explain the repair path.

Key Concepts

- Step 1: Confirm that abnormal loss exists.
- Step 2: Decide whether clues point toward structure, plumbing, equipment, or multiple issues.
- Step 3: Isolate lines, fittings, features, and suspected shell penetrations.
- Step 4: Verify with a second method when possible, especially before excavation.
- Step 5: Explain what was tested, what was ruled out, what remains likely, and what should happen next.

Diagnostic Notes

- A test that rules something out is still valuable.
- Use the customer's story, but do not let it trap the diagnosis.
- Confidence comes from sequence, not bravado.

Module Knowledge Check

M4. What is the safest technical framing for leak detection?

- A.** A systematic diagnostic process
- B.** A quick guess based on the first visible crack
- C.** A repair-first trade where testing is optional

Answer: _____

Dye Testing

Dye testing is useful when you already have a suspected visible leak point. Still water matters because the technician is watching whether dye gets pulled into a gap.

Technical takeaway

Dye testing helps visually confirm suspected leak points around skimmers, lights, fittings, cracks, and other accessible openings.

Key Concepts

- Best use: visible or reachable features such as skimmers, lights, returns, steps, cracks, and fittings.
- Poor use: broad unexplained water loss with no suspected area, moving water, wind, or hidden underground plumbing.
- A positive result looks like dye being pulled into a specific opening rather than drifting randomly.
- Dye can support a repair recommendation, but weak or confusing movement may require another test.
- Pool lights involve electrical safety. Any inspection or repair should follow trained electrical-safety procedure.

Diagnostic Notes

- Dye confirms a suspected location; it does not magically search the whole pool.
- Turn circulation off and let water get still before testing when procedure allows.
- Use dye as evidence, not theater.

Module Knowledge Check

M5. Where is dye testing most useful?

- A.** At visible suspected leak points
- B.** Only at the equipment pad
- C.** Inside every underground pipe

Answer: _____

Pressure Testing

Pressure testing isolates pool plumbing lines. If a sealed line loses pressure, the technician has evidence that the line may be compromised.

Technical takeaway

Pressure testing isolates plumbing issues by sealing a line, applying controlled pressure, and watching whether it holds.

Key Concepts

- Lines are isolated with plugs so the technician can test one section at a time.
- A gauge that holds steady suggests the line is likely intact under that test condition.
- A pressure drop suggests a possible leak in that isolated line or test setup.
- Results should be interpreted with training because valves, plugs, air, water, and setup details matter.
- Pressure testing is not casual DIY work. It requires proper equipment, procedure, and safety judgment.

Diagnostic Notes

- Use pressure testing to distinguish plumbing from structure.
- A failed test narrows the search; it does not always pinpoint the exact buried break by itself.
- Verification matters before recommending excavation.

Module Knowledge Check

M6. What does a pressure drop usually indicate during an isolated line test?

- A. The isolated line or setup is not holding pressure
- B. The pool is only evaporating
- C. The skimmer basket is full

Answer: _____

Common Leak Locations

You do not need to memorize every rare failure. Start with the common areas and learn why they fail.

Technical takeaway

Common leak areas include skimmers, lights, return fittings, visible cracks, main drains, and the equipment pad.

Key Concepts

- Skimmers are common because plastic, concrete, plaster, and movement meet at the throat.
- Pool lights are common because the niche, conduit, and sealing points create paths for water.
- Return fittings can leak around the fitting face or in the buried return line.
- Expansion cracks and shell cracks matter, especially in older pools or after movement.
- The equipment pad can create a false pool leak if water is visibly leaking from pumps, filters, heaters, valves, or unions.

Diagnostic Notes

- Common does not mean certain.
- Water level clues help prioritize features.
- Check the equipment pad before chasing hidden problems.

Module Knowledge Check

M7. Which is a common leak area because materials meet and separate?

- A. Fence gate
- B. Pool thermometer
- C. Skimmer throat

Answer: _____

Concrete and Guniting Shell Repair Basics

Concrete, guniting, and shotcrete pools have a structural shell under the interior finish. Repair thinking starts by deciding whether the issue is structural shell movement, weak material, or only surface finish distress.

Technical takeaway

Concrete or guniting repair starts with diagnosis: determine whether the defect is a surface finish issue, a bond failure, weak shell material, or a structural crack.

Key Concepts

- Guniting and shotcrete are pneumatically placed concrete materials used to form pool shells over steel reinforcement.
- Shell problems can show up as open cracks, recurring leaks, hollow-sounding areas, exposed reinforcement, rust staining, or movement around steps, benches, bond beams, and fittings.
- Sounding is a common concept: solid areas tend to ring differently than hollow, weak, or debonded areas when checked by a trained technician.
- Weak or unsuitable material should not be buried under a cosmetic patch. Sound repair preparation means removing loose or deteriorated material and reaching sound substrate.
- Structural crack repair is not the same as smearing sealant over a crack. It may require engineering judgment, staples, injection systems, waterproofing details, or shell reconstruction depending on severity.

Diagnostic Notes

- Separate cosmetic symptom from structural cause.
- A leak at a crack should trigger both leak thinking and structural thinking.
- The procedure-safe answer is: I would follow approved procedure and escalate structural concerns rather than freelancing a repair.

Module Knowledge Check

M8. What is the key distinction in concrete or guniting repair thinking?

- A. Whether the pool has a ladder
- B. Blue pool versus green pool
- C. Surface finish issue versus shell/structural issue

Answer: _____

Pool Finish and Plaster Repair Basics

The finish is the visible interior surface: plaster, quartz, pebble, exposed aggregate, or similar cementitious coatings. Finish repair depends on bond, chemistry, prep, cure, color expectations, and whether the damage is local or widespread.

Technical takeaway

Pool finish repair depends on bond, surface preparation, compatible material, curing/startup, and whether the defect is isolated or widespread.

Key Concepts

- Common finish issues include check cracks, open cracks, etching, scaling, stains, blisters, spalls, delamination, hollow spots, and roughness.
- Delamination means the finish has separated within itself or from the substrate. Small isolated areas may be patched; widespread hollow areas often point toward resurfacing evaluation.
- A durable patch is about preparation, not just material. The repair area should be clean, sound, properly roughened, and compatible with the coating system.
- Color and texture matching are hard. A good customer explanation sets expectations that a patch may stop a defect without becoming invisible.
- Fresh plaster needs careful startup: continuous fill, brushing, balanced water, plaster dust control, and a roughly 28-day early-care window.

Diagnostic Notes

- Do not promise a perfect color match on aged finish.
- Ask whether the issue is isolated or widespread before recommending patch versus resurfacing.
- Connect repair quality to prep, bond, cure, and water chemistry.

Module Knowledge Check

M9. Why is a plaster patch not just a cosmetic smear?

- A. Any patch automatically matches forever
- B. Water chemistry does not matter
- C. Bond, prep, material compatibility, and curing affect whether it lasts

Answer: _____

Communicating Technical Findings

Technical findings should be communicated clearly and specifically. The goal is to explain the evidence, the likely cause, and the repair path without vague conclusions.

Technical takeaway

A strong technical explanation names what was tested, what was observed, what it indicates, and what should happen next.

Key Concepts

- Clear documentation reduces confusion and supports better repair decisions.
- Specifics matter: name the test, the result, and the next step.
- Avoid vague conclusions like, 'Repairs are needed.' Explain the evidence.
- Distinguish confirmed findings from likely findings.
- A good report turns field observations into an understandable repair recommendation.

Diagnostic Notes

- Start with the water loss or defect being investigated.
- Translate test results into plain language.
- Use certainty carefully. Separate what is confirmed from what requires further evaluation.

Module Knowledge Check

M10. Which technical explanation is strongest?

- A.** The pool is bad.
- B.** The return line was pressure tested, pressure loss was observed, and the next step is locating and repairing that line.
- C.** Repairs are needed.

Answer: _____

Documentation and Repair Recommendations

A complete leak detection workflow connects evidence to documentation and a practical repair recommendation. The technical record should make the next step clear.

Technical takeaway

A strong workflow moves from observed concern to test evidence, likely cause, recommended repair path, and follow-up verification.

Key Concepts

- A complete service record captures the concern, conditions, tests performed, results, and recommended next step.
- Repair recommendations should be tied to what was tested and observed.
- Photos, measurements, pressure readings, dye observations, and water-level clues can support the technical record.
- If a condition may involve structural movement, electrical safety, excavation, or diving, it should be handled under trained procedure.
- Follow-up verification matters after repair because the original water loss may have had more than one cause.

Diagnostic Notes

- Document what was ruled in and what was ruled out.
- Do not let a repair recommendation outrun the evidence.
- A clear report should let another trained person understand the diagnostic path.

Module Knowledge Check

M11. What should a repair recommendation be tied to?

- A. Only sales pressure
- B. Diagnosis, test evidence, observed conditions, and follow-up verification
- C. Only swimming ability

Answer: _____

Technical Vocabulary and Field Review

This review consolidates the technical vocabulary used throughout the course. The goal is accurate terminology and clear distinctions between systems, tests, and repair categories.

Technical takeaway

Use technical vocabulary to make the diagnosis clearer: suction side, pressure side, structural leak, plumbing leak, equipment leak, shell defect, and finish defect.

Key Concepts

- Core leak terms: leak detection, pressure testing, dye testing, plumbing isolation, return line, suction line, skimmer, main drain, equipment pad, structural leak, evaporation loss.
- Core repair terms: gunite, shotcrete, shell, substrate, delamination, spall, bond failure, plaster, finish, patch, resurfacing, and startup.
- Suction-side issues occur before the pump; pressure-side issues occur after the pump.
- Structural/shell issues and finish/plaster issues can look similar at first, but they often require different repair paths.
- A useful technical explanation connects symptom, test, finding, and recommendation.

Diagnostic Notes

- Use terminology only when it makes the explanation more precise.
- Separate leak detection language from repair language.
- Review the diagnostic sequence before drawing a conclusion.

Module Knowledge Check

M12. Which statement best summarizes the technical workflow?

- A. Ignore documentation once a leak is suspected.
- B. Guess based on the first visible mark.
- C. Confirm the symptom, test likely causes, document the finding, and recommend the next technical step.

Answer: _____

Vocabulary Glossary

Term	Plain-English meaning
Pressure testing	Sealing and pressurizing an isolated plumbing line to see whether it holds pressure.
Dye testing	Using leak-detection dye near a suspected visible leak point to watch whether water pulls dye into an opening.
Suction line	Plumbing before the pump that pulls water from skimmers or main drains.
Return line	Plumbing after the filter/heater that sends water back into the pool through returns.
Structural leak	A leak in the pool shell, skimmer area, light niche, fitting, crack, or other pool structure.
Equipment pad	The area where pump, filter, heater, valves, unions, and related equipment are located.
Evaporation loss	Water lost naturally to weather conditions rather than through a defect.
Plumbing isolation	Separating one line or system section so it can be tested without confusing results.
Gunite	A pneumatically applied concrete material commonly used to form reinforced pool shells.
Delamination	A bond or layer failure where finish material separates from itself or the substrate.
Spall	A lifted or broken-away upper finish layer, often related to bond, impact, drying, or chemistry issues.
Startup	The early care period after plastering when fill, brushing, water balance, and dust control protect the finish.

Final Knowledge Check

The choices below are randomized for this PDF version. Use the answer key only after a full attempt.

Q1. Which path best describes normal pool circulation?

- A. Pool to returns to filter to skimmer to pump
- B. Pump to pool to skimmer to drain to heater
- C. Pool to skimmer/drain to pump to filter to heater to returns

Answer: _____

Q2. Why can an autofill system make a leak harder to notice?

- A. It permanently seals small plumbing leaks
- B. It replaces lost water and can hide the true loss rate
- C. It makes dye testing unnecessary

Answer: _____

Q3. In a bucket test, what does it mean if the pool drops more than the bucket?

- A. Normal evaporation only
- B. The filter is definitely broken
- C. Likely leak beyond evaporation

Answer: _____

Q4. What is the safest technical framing for leak detection?

- A. A repair-first trade where testing is optional
- B. A systematic diagnostic process
- C. A quick guess based on the first visible crack

Answer: _____

Q5. Where is dye testing most useful?

- A. Inside every underground pipe
- B. Only at the equipment pad
- C. At visible suspected leak points

Answer: _____

Q6. What does a pressure drop usually indicate during an isolated line test?

- A. The skimmer basket is full
- B. The pool is only evaporating
- C. The isolated line or setup is not holding pressure

Answer: _____

Q7. Which is a common leak area because materials meet and separate?

- A. Skimmer throat
- B. Pool thermometer
- C. Fence gate

Answer: _____

Q8. What is the key distinction in concrete or gunite repair thinking?

- A. Surface finish issue versus shell/structural issue
- B. Whether the pool has a ladder
- C. Blue pool versus green pool

Answer: _____

Q9. Why is a plaster patch not just a cosmetic smear?

- A. Any patch automatically matches forever
- B. Water chemistry does not matter
- C. Bond, prep, material compatibility, and curing affect whether it lasts

Answer: _____

Q10. Which technical explanation is strongest?

- A. The pool is bad.
- B. The return line was pressure tested, pressure loss was observed, and the next step is locating and repairing that line.
- C. Repairs are needed.

Answer: _____

Q11. What should a repair recommendation be tied to?

- A. Only swimming ability
- B. Only sales pressure
- C. Diagnosis, test evidence, observed conditions, and follow-up verification

Answer: _____

Q12. Which statement best summarizes the technical workflow?

- A. Guess based on the first visible mark.
- B. Confirm the symptom, test likely causes, document the finding, and recommend the next technical step.
- C. Ignore documentation once a leak is suspected.

Answer: _____

Answer Key

Module Knowledge Checks

M1. B - Pool to skimmer/drain to pump to filter to heater to returns

Water is pulled from the pool by skimmers/main drains, moved by the pump, filtered, optionally heated, and returned.

M2. A - It replaces lost water and can hide the true loss rate

Autofill keeps the level up, so the customer may not see how much water is being lost.

M3. B - Likely leak beyond evaporation

The bucket accounts for evaporation. Extra pool drop points to water leaving somewhere else.

M4. A - A systematic diagnostic process

The strongest signal is that you understand methodical diagnosis.

M5. A - At visible suspected leak points

Dye is visual. It works best where a technician can place dye near a suspected opening.

M6. A - The isolated line or setup is not holding pressure

A dropping gauge means pressure is escaping somewhere in that isolated test path or setup.

M7. C - Skimmer throat

Skimmer throats are classic suspects because the skimmer and shell can separate.

M8. C - Surface finish issue versus shell/structural issue

The repair path depends on whether the defect is in the finish layer, the bond, or the structural shell.

M9. C - Bond, prep, material compatibility, and curing affect whether it lasts

Finish repairs depend on preparation, bond, compatible materials, curing/startup, and later water balance.

M10. B - The return line was pressure tested, pressure loss was observed, and the next step is locating and repairing that line.

The strongest explanation names the test, result, and next step.

M11. B - Diagnosis, test evidence, observed conditions, and follow-up verification

A technical recommendation should be grounded in observed conditions and test evidence.

M12. C - Confirm the symptom, test likely causes, document the finding, and recommend the next technical step.

The technical workflow connects confirmation, testing, documentation, and recommendation.

Final Knowledge Check

Q1. C - Pool to skimmer/drain to pump to filter to heater to returns

Water is pulled from the pool by skimmers/main drains, moved by the pump, filtered, optionally heated, and returned.

Q2. B - It replaces lost water and can hide the true loss rate

Autofill keeps the level up, so the customer may not see how much water is being lost.

Q3. C - Likely leak beyond evaporation

The bucket accounts for evaporation. Extra pool drop points to water leaving somewhere else.

Q4. B - A systematic diagnostic process

The strongest signal is that you understand methodical diagnosis.

Q5. C - At visible suspected leak points

Dye is visual. It works best where a technician can place dye near a suspected opening.

Q6. C - The isolated line or setup is not holding pressure

A dropping gauge means pressure is escaping somewhere in that isolated test path or setup.

Q7. A - Skimmer throat

Skimmer throats are classic suspects because the skimmer and shell can separate.

Q8. A - Surface finish issue versus shell/structural issue

The repair path depends on whether the defect is in the finish layer, the bond, or the structural shell.

Q9. C - Bond, prep, material compatibility, and curing affect whether it lasts

Finish repairs depend on preparation, bond, compatible materials, curing/startup, and later water balance.

Q10. B - The return line was pressure tested, pressure loss was observed, and the next step is locating and repairing that line.

The strongest explanation names the test, result, and next step.

Q11. C - Diagnosis, test evidence, observed conditions, and follow-up verification

A technical recommendation should be grounded in observed conditions and test evidence.

Q12. B - Confirm the symptom, test likely causes, document the finding, and recommend the next technical step.

The technical workflow connects confirmation, testing, documentation, and recommendation.

References

American Leak Detection: Bucket Test

<https://www.americanleakdetection.com/diy-leak-tests/bucket-test/>

Bucket test procedure and interpretation.

Anderson Manufacturing: Pressure Test Plumbing

<https://www.leaktools.com/pressure-test-plumbing.html>

Pressure testing as a process-of-elimination method for structure versus plumbing leaks.

National Plasterers Council: Pool Plaster FAQ

<https://www.nationalplastererscouncil.com/faq/>

Generic plaster and finish concepts including cracking, bonding, delamination, spalls, and repair judgment.

Pool Engineering: Minor Guniting or Shotcrete Repair

<https://www.pooleng.com/procedures-for-minor-guniting-or-shotcrete-repair-in-swimming-pools-2/>

Generic shell repair concepts including sounding, removing weak material, roughening, cleaning, and bonding.

NPT: Pool Finish Startup Procedure

<https://www.nptpool.com/pool-finishes/start-up-procedure/>

Generic startup concepts after a new plaster finish.

Appendix: Customer Education Framework

The purpose of customer-oriented framing is not to simplify the facts until they are vague. It is to explain the diagnostic sequence clearly enough that the customer understands what was checked, what the result means, and why the next step follows from the evidence.

Core model

Concern observed -> Test performed -> Result seen -> Meaning -> Next technical step

Four-Part Explanation Pattern

- Start with the concern: water loss, visible wet area, air bubbles, a crack, hollow finish, staining, or equipment leakage.
- Name the diagnostic step: bucket test, dye test, pressure test, equipment inspection, sounding, or visual inspection.
- Translate the result: explain whether the result points toward evaporation, plumbing, structure, finish, or equipment.
- State the next step: repair, further isolation, structural evaluation, finish repair, resurfacing evaluation, or post-repair verification.

Customer-Friendly Test Explanations

Topic	Plain-language explanation
Bucket test	This compares the pool against a bucket of the same water. If the pool drops more than the bucket, the extra loss points toward a leak instead of normal evaporation.
Dye test	Dye is placed near a suspected opening while the water is still. If the dye pulls into that spot, it helps confirm where water is leaving.
Pressure test	A plumbing line is isolated and placed under controlled pressure. If the line does not hold pressure, that line or test setup needs further locating and repair.
Shell versus finish	The shell is the structural concrete or gunite body. The finish is the visible plaster, quartz, pebble, or coating. The repair path changes depending on which layer has failed.

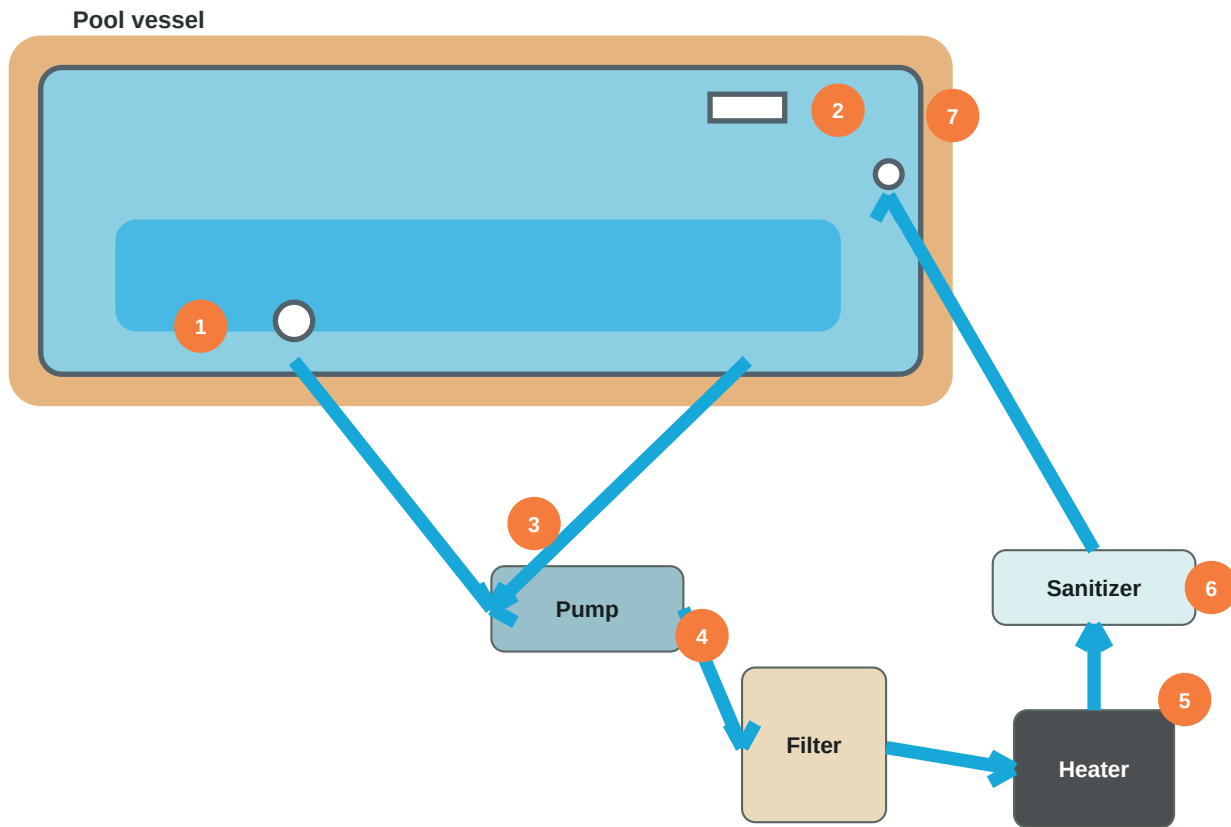
Customer Education Scripts

These are technical explanation templates. They should be adjusted to match the actual field evidence and approved service procedures.

Scenario	Educational customer framing
Likely evaporation	The pool and bucket lost about the same amount of water, so the result is consistent with evaporation under the current conditions. I would keep monitoring, especially if weather or usage changes.
Likely structural leak	The dye movement at this fitting/crack shows water pulling into that opening. That gives us a specific area to repair or evaluate further instead of guessing across the whole pool.
Likely plumbing leak	This isolated line did not hold pressure. That means the next technical step is locating where that line is losing pressure before discussing excavation or repair scope.
Finish delamination	This area sounds hollow or has separated from the surface below it. A spot patch may address an isolated area, but widespread separation usually requires a broader resurfacing evaluation.
Patch expectations	A patch is intended to correct the defect, but aged finishes rarely match perfectly in color and texture. The repair goal and appearance expectation should be discussed separately.
Post-repair verification	After the repair, the pool should be checked again because one confirmed leak does not always rule out a second source of water loss.
Avoid vague framing	Replace “You need repairs” with “Here is what was tested, here is what the result showed, and here is the next technical step.”

Appendix: Pool Circulation Anatomy

Original schematic showing the typical water path and component sequence.



Component legend

- | | |
|--------------|---------------------|
| 1 Main drain | 5 Heater |
| 2 Skimmer | 6 Sanitation system |
| 3 Pump | 7 Return lines |
| 4 Filter | |

How to read the system

- Suction side: main drain and skimmer pull water toward the pump.
- Pressure side: the pump pushes water through filtration, treatment, heat, and back to the pool.
- Leak clues often depend on where water loss or air entry appears in this sequence.

Plain-language framing

Water leaves the pool through suction points, gets cleaned and treated, then returns through jets. A diagnostic test tries to isolate which part of that path is losing water, taking in air, or failing.

Appendix: Construction and Layout Basics

Original quick reference for connecting pool construction details to leak diagnosis.

Core idea

A leak clue is easier to interpret when you know the pool type, shell material, finish, and plumbing layout.

Common pool construction types

Type	What it means	Leak/repair clues
Concrete / gunitite	Reinforced shell with plaster, quartz, pebble, or similar finish.	Cracks, penetrations, bond beam, hollow finish, delamination, fittings.
Fiberglass	Factory-made shell set into prepared excavation and backfill.	Fittings, shell movement, plumbing connections, settlement, gelcoat damage.
Vinyl liner	Framed or walled pool with replaceable liner and faceplate fittings.	Liner tears, seams, track, skimmer/return faceplates, steps.
Above-ground	Wall and liner system with exposed equipment and shorter plumbing runs.	Wall corrosion, liner punctures, skimmer/return gaskets, visible equipment leaks.

Plan-reading checklist

- Identify suction points: skimmers, main drains, dedicated cleaner lines, and valves.
- Identify pressure-side returns, water features, heaters, sanitizers, and valve paths.
- Locate penetrations through the shell: lights, fittings, drains, skimmers, steps, and niches.
- Note deck drains, expansion joints, settlement, drainage, and areas where water may travel.
- Connect each visible symptom to the nearest component or plumbing route before recommending repair.

Customer framing

The pool's construction tells us where leaks are most likely and what kind of repair makes sense. A concrete shell crack, vinyl liner tear, fiberglass fitting leak, and equipment leak can all lose water, but each one calls for a different diagnostic path and repair expectation.

Appendix: Electrical Safety Awareness

Conceptual safety reference for pools, lights, equipment, bonding, grounding, and escalation.

Stop-and-escalate rule

Water and electricity are a high-risk combination. Pool lights, junction boxes, bonding conductors, outlets, heaters, automation, and damaged wiring should be handled only under approved electrical-safety procedure by qualified personnel.

Terms to understand

Concept	Plain-language meaning	Why it matters
Bonding	Connecting conductive pool parts together so voltage differences are reduced.	A bonding problem can create shock risk even when equipment appears to run.
Grounding	Providing a fault path back to the electrical system grounding point.	Grounding helps protective devices operate during faults.
GFCI protection	Ground-fault protection that trips when current leaks from the intended path.	Trips are safety clues, not annoyances to ignore.
Junction boxes	Electrical connection points for pool lights and related wiring.	Water intrusion, damage, or improper covers require qualified evaluation.
Pool lights	Fixtures installed in or near the pool shell.	Leak work near lights must respect electrical isolation, conduit paths, and approved procedures.

Field awareness checklist

- Do not remove pool lights, covers, or junction-box components unless trained and authorized.
- Treat tripped breakers, GFCI trips, tingling sensations, corrosion, heat, or water in electrical areas as safety flags.
- Before leak or repair work near lights/equipment, confirm the approved isolation and lockout procedure.
- Keep customer explanations simple: electrical safety must be resolved before leak repair proceeds in that area.

Customer framing

This area involves electrical safety, so the safe next step is qualified evaluation before continuing leak repair work around that component.